

AGAIN Light is the great informant + some Physics too = equations to
Use for our observations. = **STELLAR PARAMETERS**

Kepler's law $M_1 + M_2 = a^3/p^2$,
 $M_2/M_1 = V_1/V_2$ and $M_2/M_1 = r_1/r_2$
a is the distance between the stars, p the orbital
Period. We can use Doppler shift or
direct measurements.

- **Mass:** Orbits of binary stars.

$$L = M^4$$

$$\text{Life : life time} = M/L = 1/M^3$$

- **Surface temperature:** Black body curve, plus some spectral information

$$T = 0.029/\lambda \quad \text{Wien's law}$$

- **Chemical composition:** Absorption lines in the spectrum.

- **Distance:** Parallax for nearby stars. Variable stars for further distances.

$$p = 1/d \quad p \text{ in parsecs, } d \text{ in arc seconds}$$

- **Luminosity:** The measured apparent brightness and the distance.

$$L = 4 \pi R^2 \sigma T^4 \quad B = L / 4 \pi d^2 \quad (\text{inverse square law})$$

- **Radius:** The luminosity and the temperature.

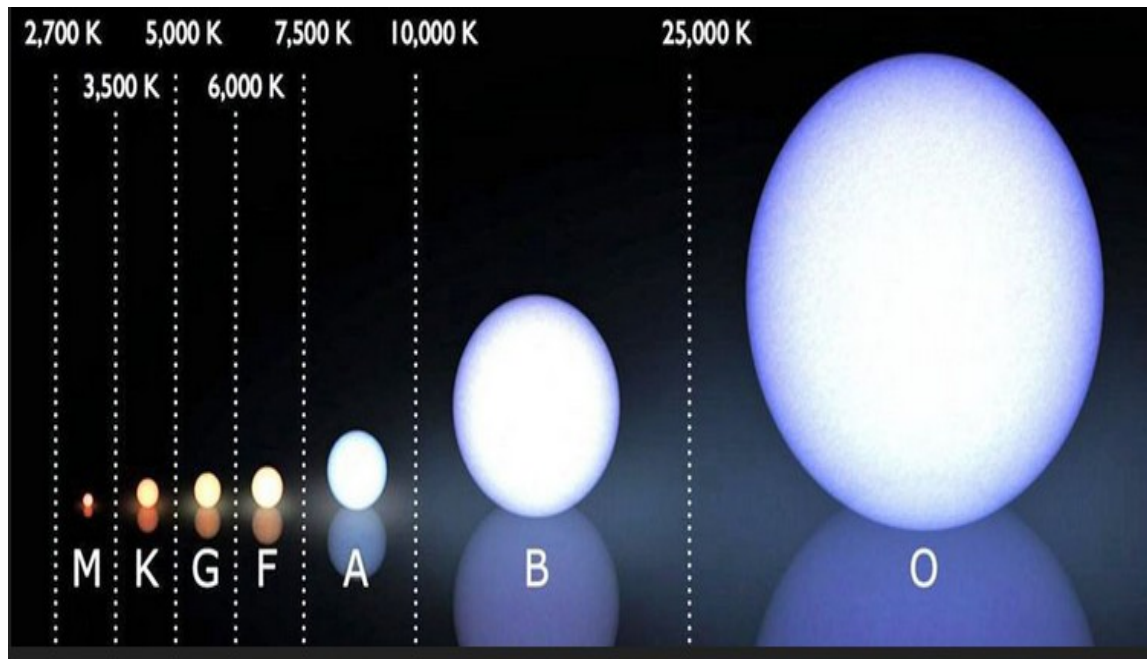
$$L = 4 \pi R^2 \sigma T^4$$

$$\text{Life : life time} = M/L = 1/M^{2.5}$$

See applet about $L = f(R, T)$

How to find the mass of the stars

All the intrinsic properties
are dictated by the mass



We use binary systems to find the mass of a few hundred stars then infer relationship between mass and other properties

Types of Binary Star Systems

- Visual binary
- Eclipsing binary
- Spectroscopic binary

About half of all stars are in binary systems.

**MASSES OF STARS WERE FOUND thanks to binary system.
Then the relationship between Luminosity and Mass was found.**

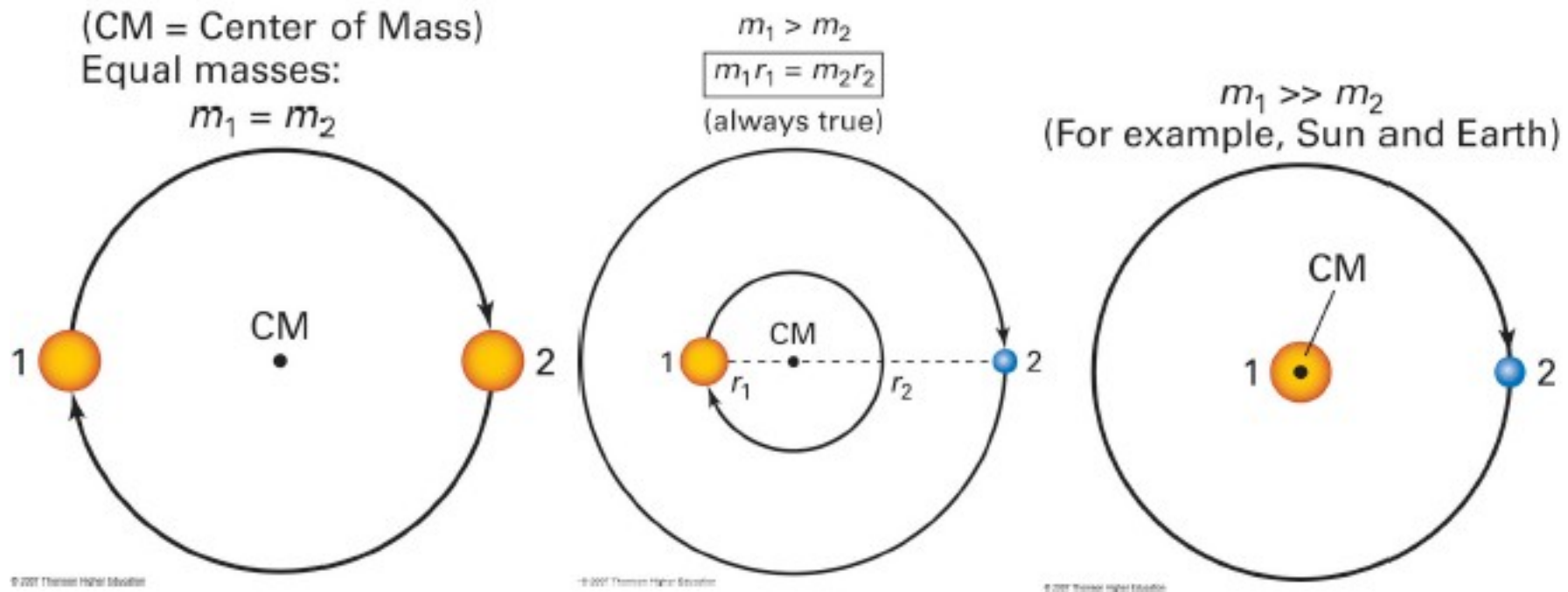


The orbit of a binary star system (distance and speed) depends on strength of gravity and therefore on the masses.

Binary Stars ("double stars")

Over 50% of the "stars" you see are actually multiple systems bound by gravity. Most are **double**. Sometimes there are even more (3, 4 or more). They go around their center of mass (CM)

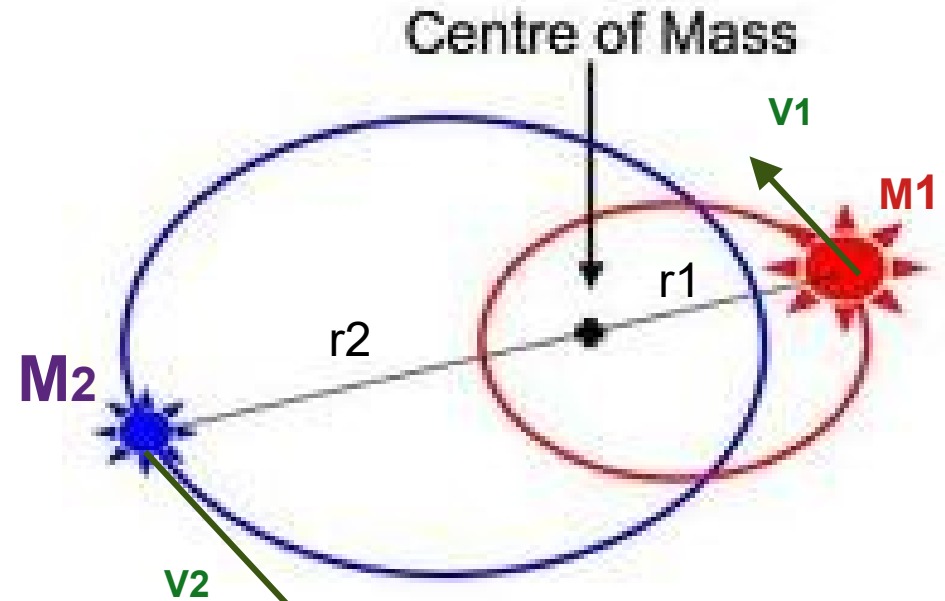
Video binary



We can use the observed orbits to derive masses, We understand how the orbits work.

See movies binaries

Center of Mass: like a see-saw

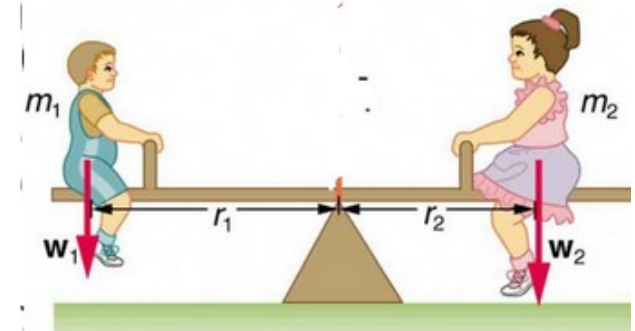


$$P^2 = \frac{a^3}{(M_1 + M_2)}$$

P in years

a = mean separation in A.U.

star masses in solar masses



They orbit their common center of mass.

$M_1 r_1 = M_2 r_2$ (seesaw principle) and $M_1 v_1 = M_2 v_2$ (conservation of momentum)

If $M_1 \gg m_2$ like planets around the Sun, the star is closed to the center of mass.

Jupiter + Sun = the center of mass is just past the Sun.

Jupiter + Earth center of mass is in the Sun.

Kepler's 3rd Law

When something is in orbit, Centripetal Force is caused by Gravitational Force.

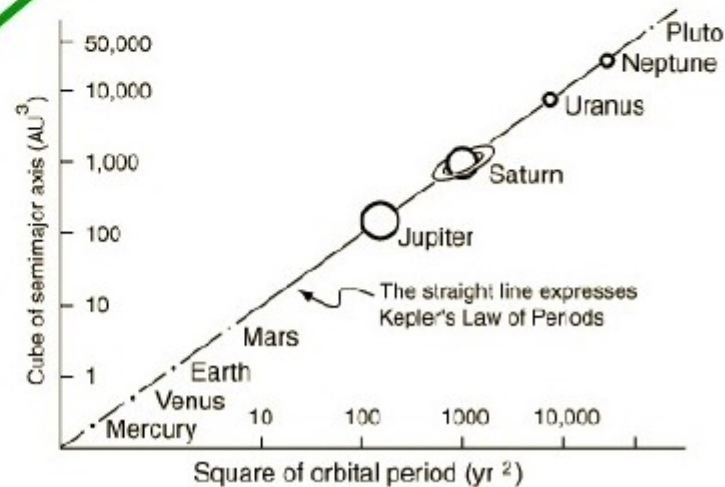
$$\frac{mv^2}{r} = G \frac{Mm}{r^2} + v = \frac{2\pi r}{T}$$

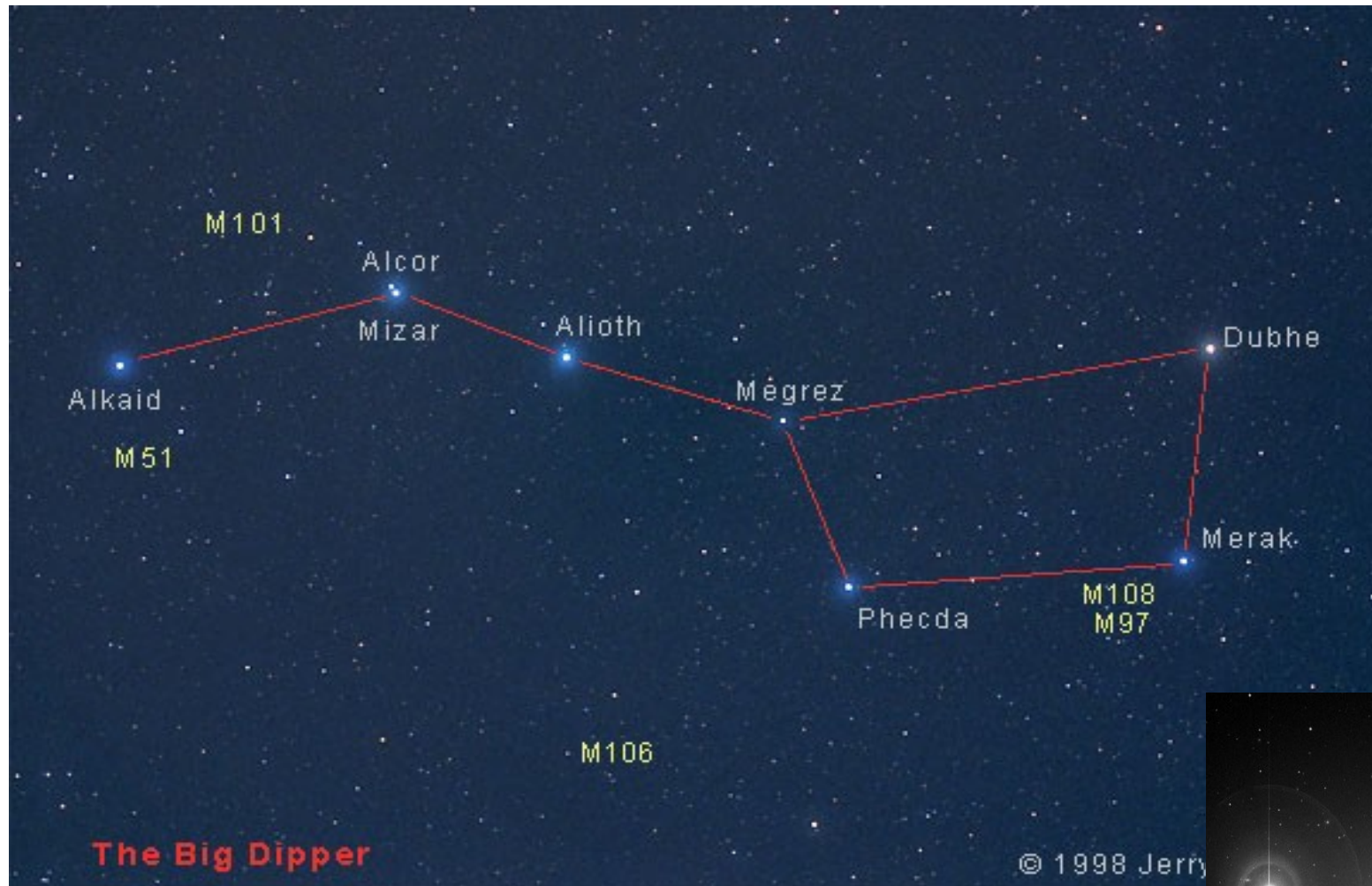
$$m \left(\frac{2\pi}{T} \right)^2 r = G \frac{Mm}{r^2}$$

$$\frac{T^2}{r^3} = \frac{4\pi^2}{GM}$$

$$T^2 \propto r^3$$

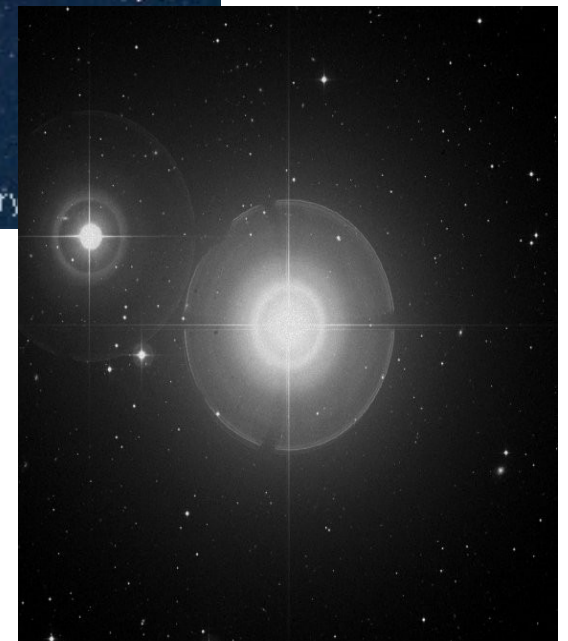
The 3rd Law: The **square of the orbital period** of a planet is **directly proportional** to the **cube of the semi-major axis** of its orbit



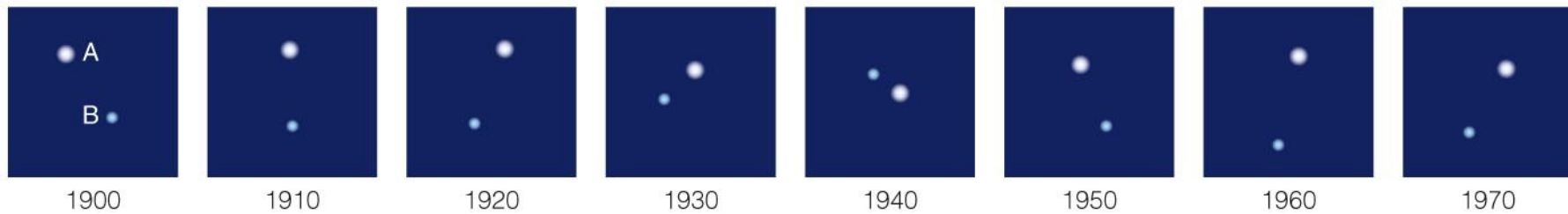


Alcor and Mizar are **optical binaries**. Fake binaries.
They are not bound gravitation-ally

Used as a way to check eyesight/



Visual Binary

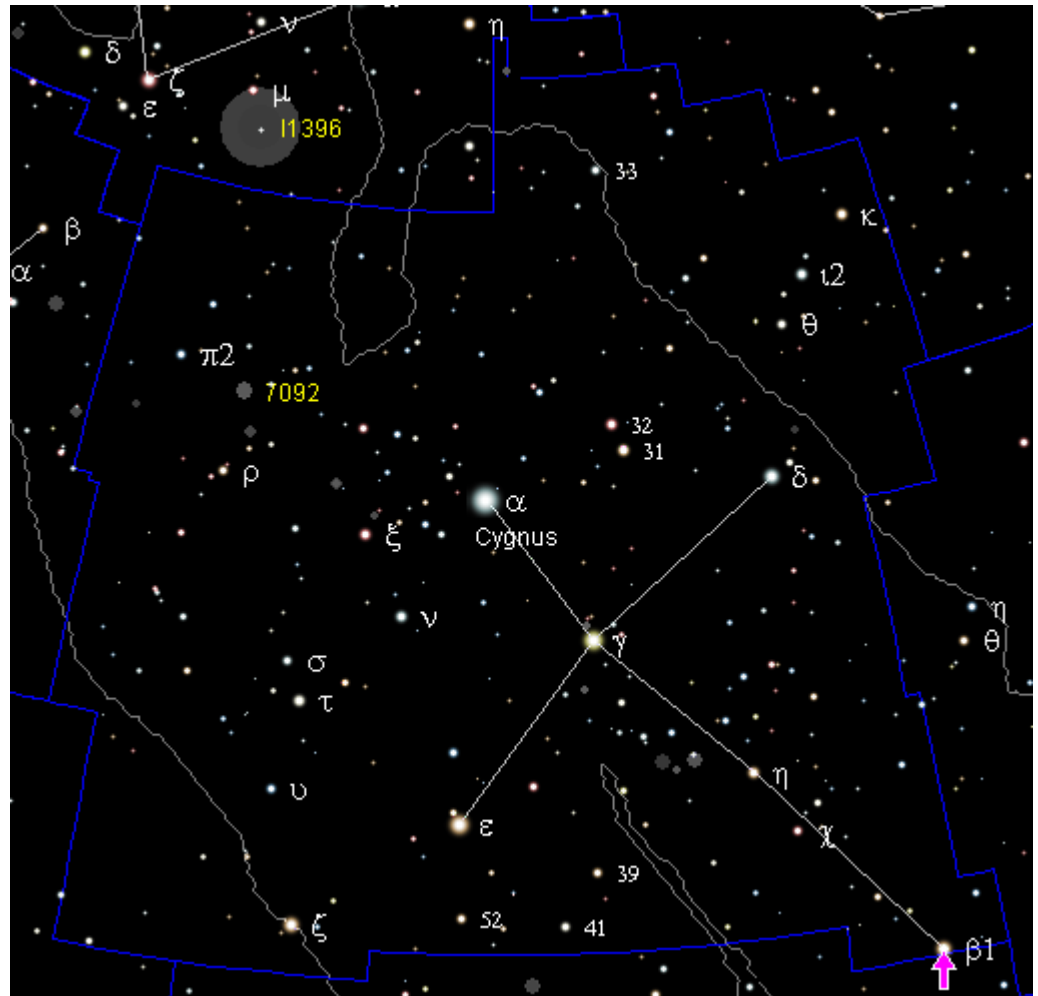


We can directly observe the orbital motions of these stars.

ALBIREO BINARY SYSTEM OR CAL STAR

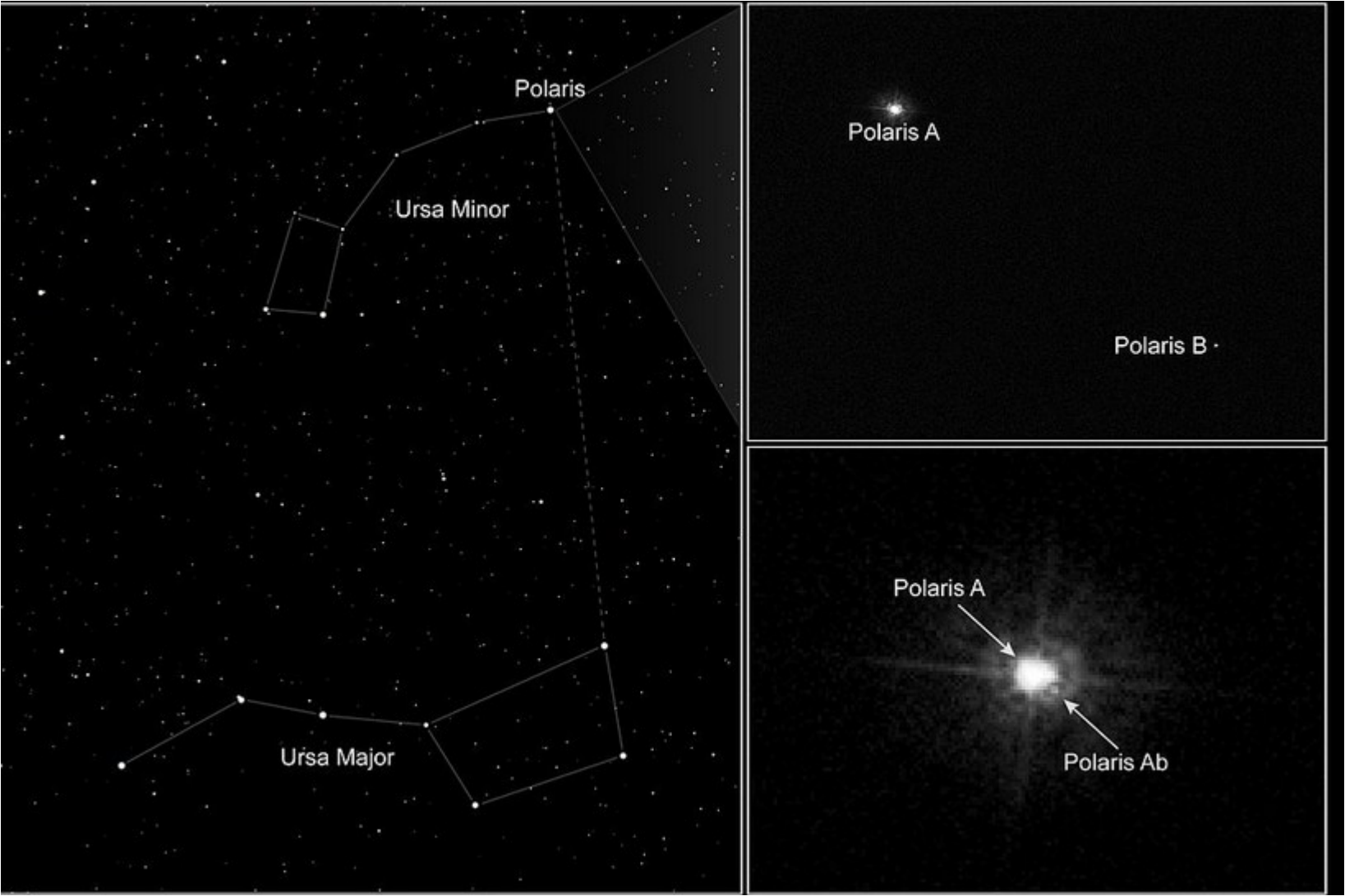


Gravitational bound together
Blue and gold. Blue star and red giant.
Beta cygnus

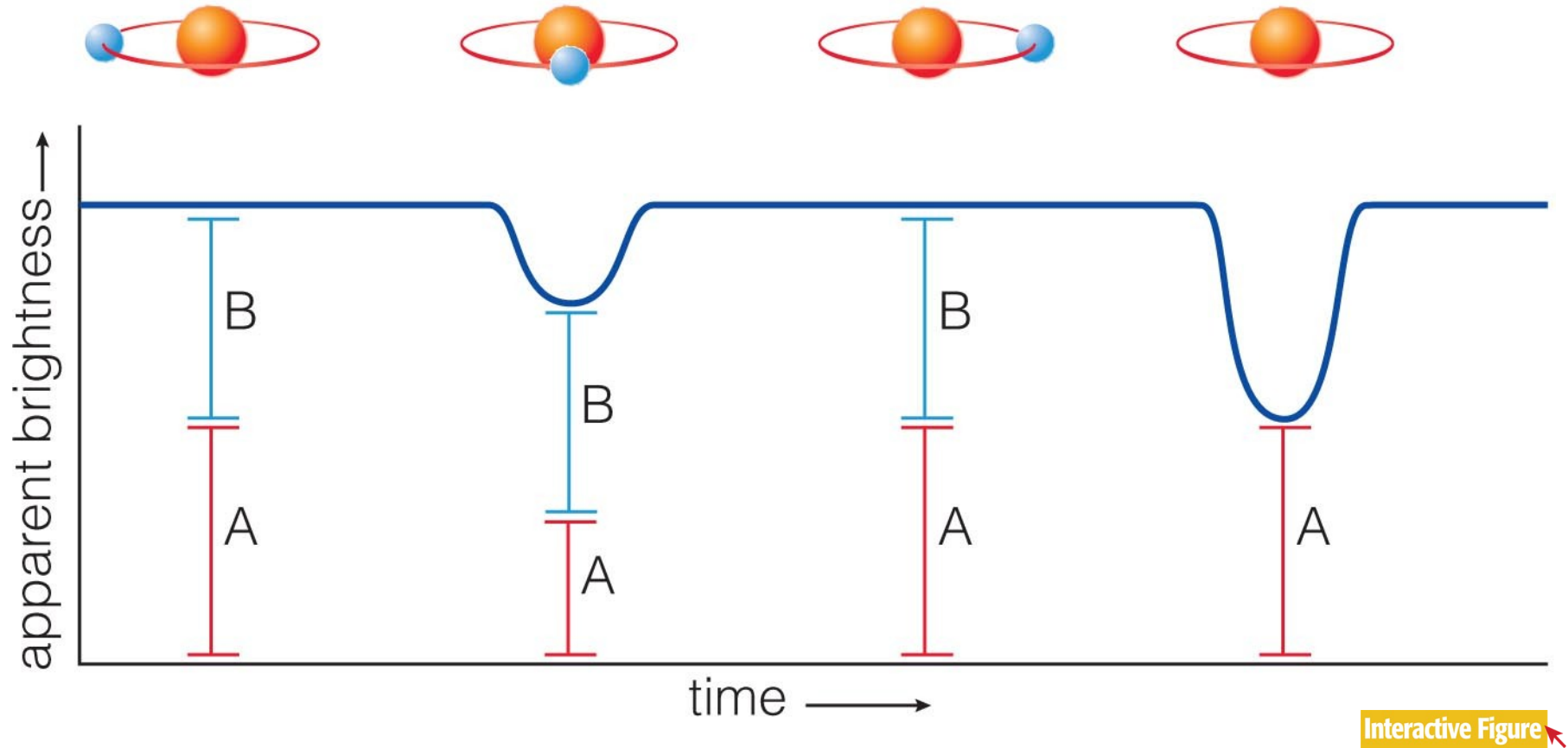


constellation
Cygnus

Star system. Polaris is a triple star system consisting of the yellow **supergiant Polaris Aa** and **two white (spectral type F) main sequence stars**, Polaris Ab and Polaris B. Polaris Aa and Ab are in close orbit with each other and Polaris B is orbiting the pair. Polaris Aa has the spectral classification F7Ib, Polaris Aa, which is a Cepheid variable



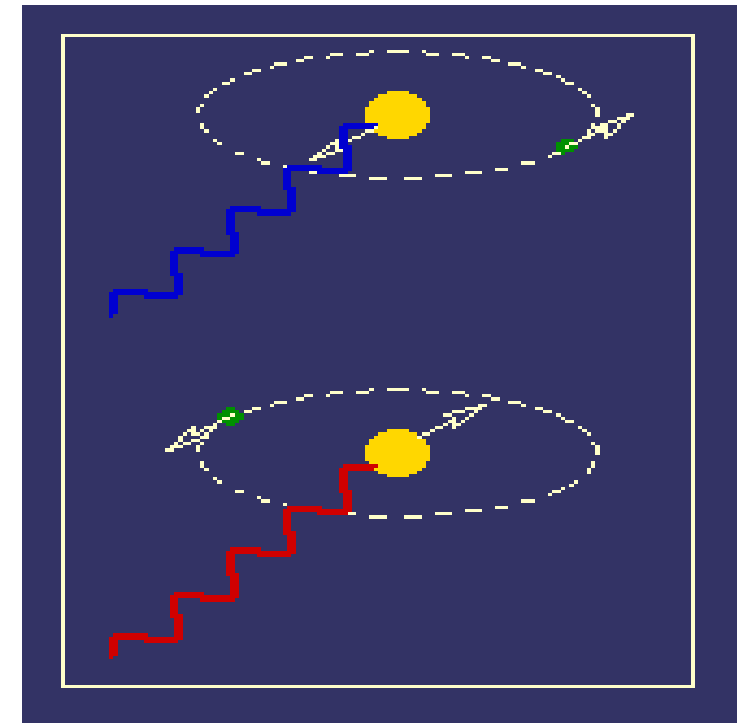
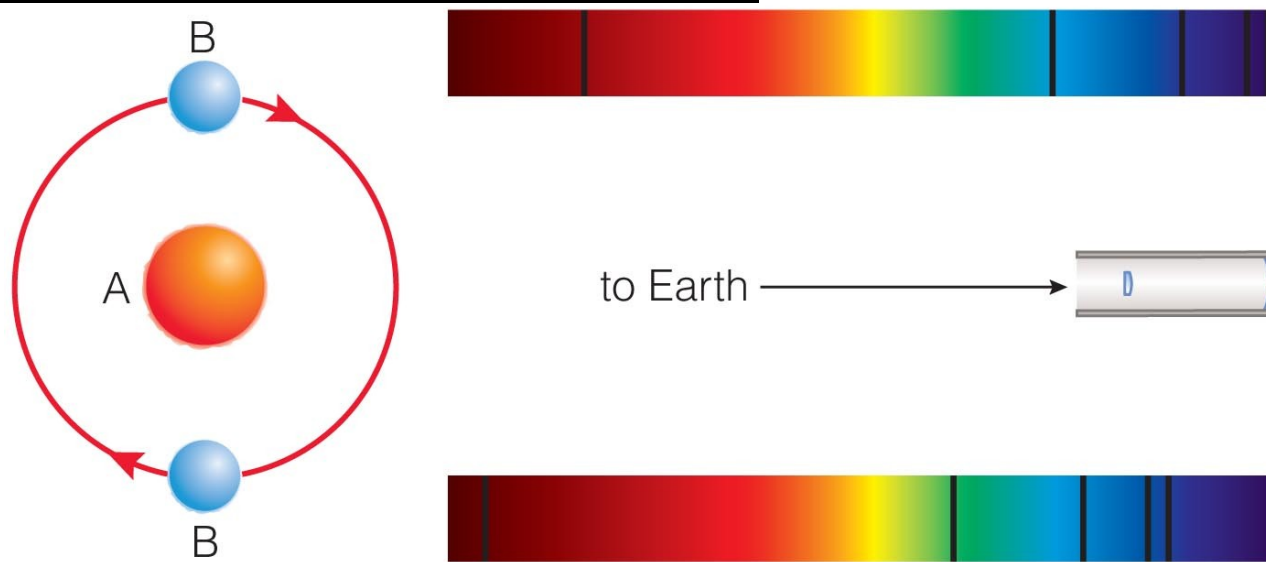
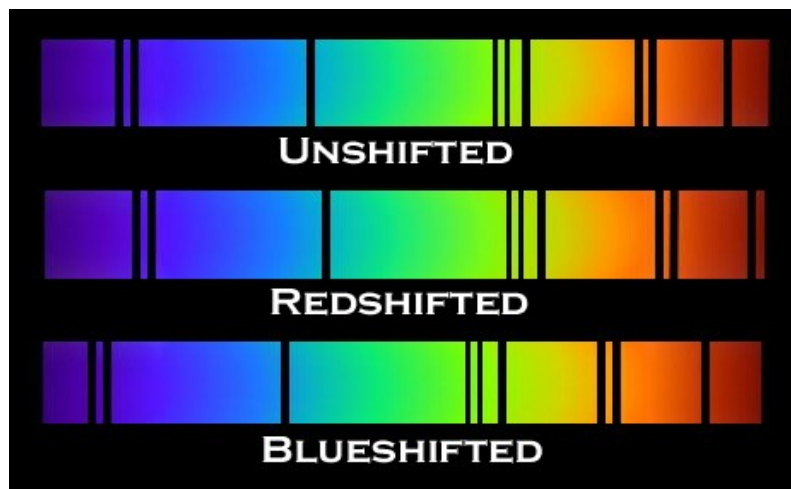
Eclipsing Binary



We can measure periodic eclipses.

Movie transit and animation

Spectroscopic Binary

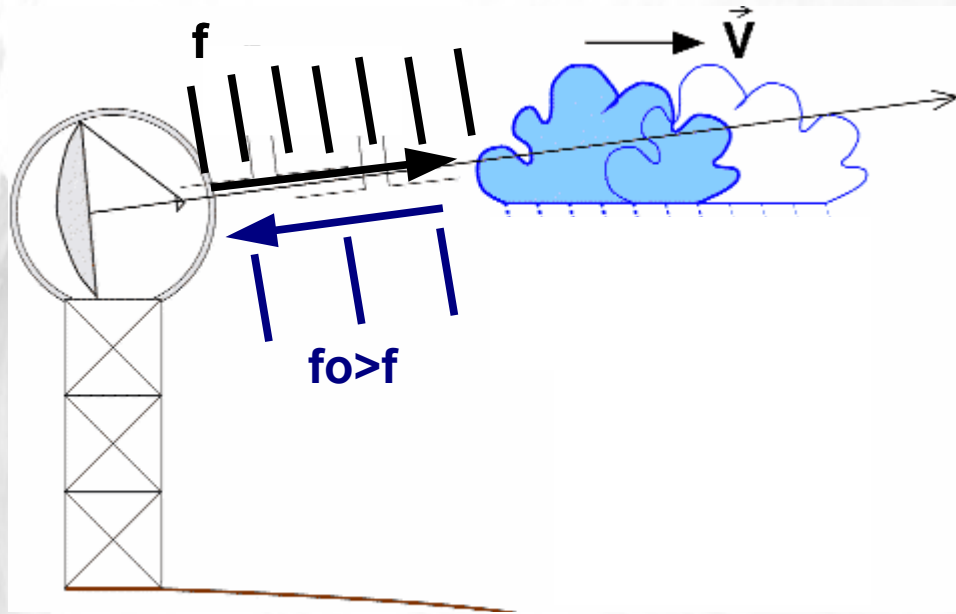


We determine the orbit by measuring Doppler shifts.

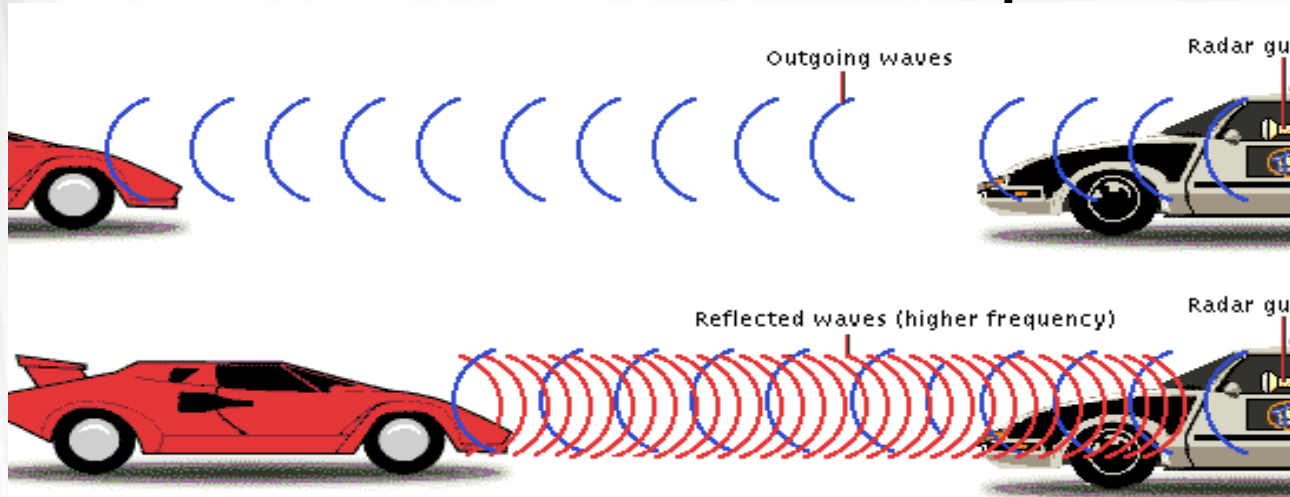
2 videos Doppler See doppler next slides

Doppler effect for waves

Incorporating the Doppler effect and echolocation makes it possible to immediately determine the speed of an approaching or departing object

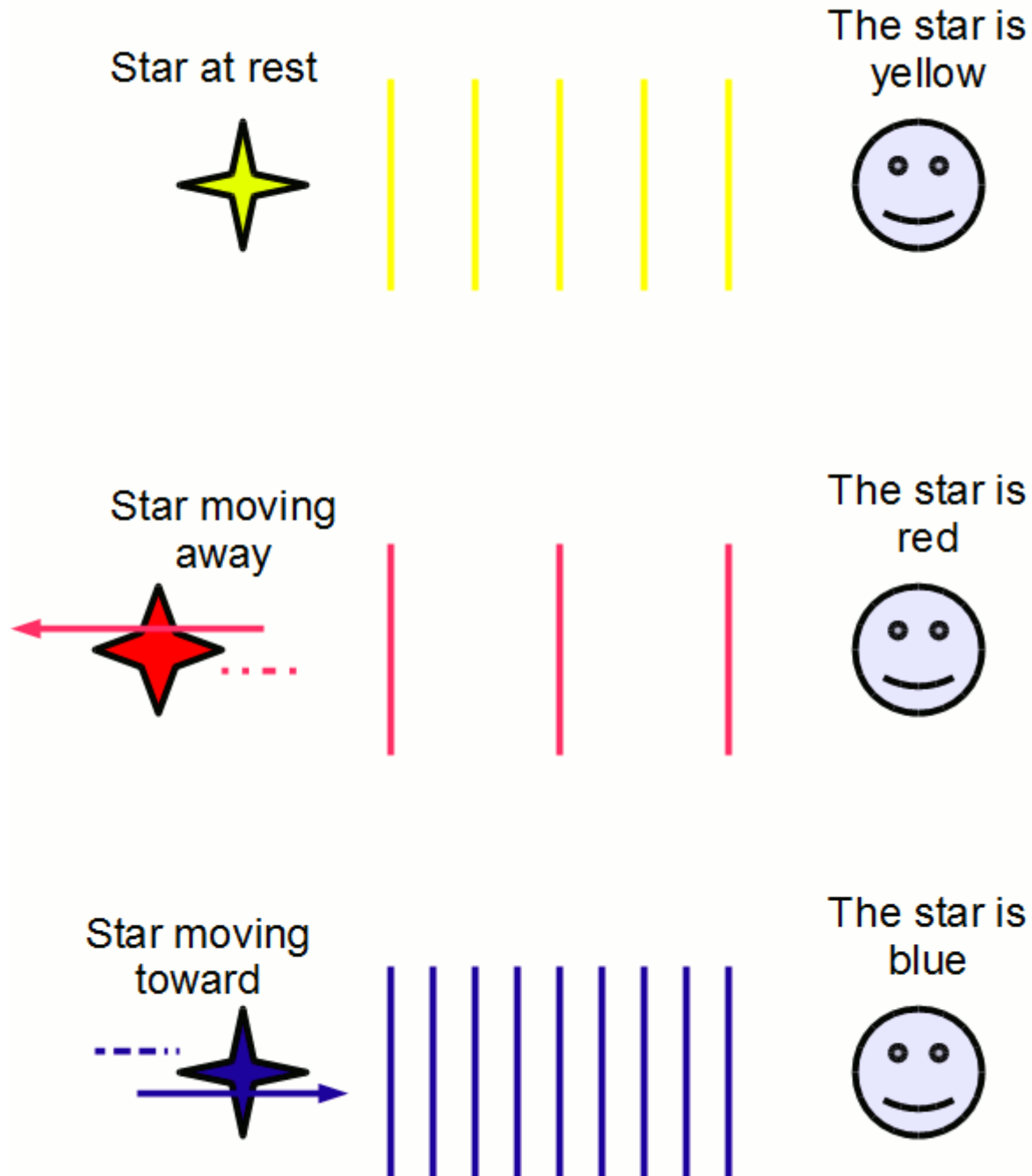


We can find the speed and location of A storm by measuring the shift in frequency.

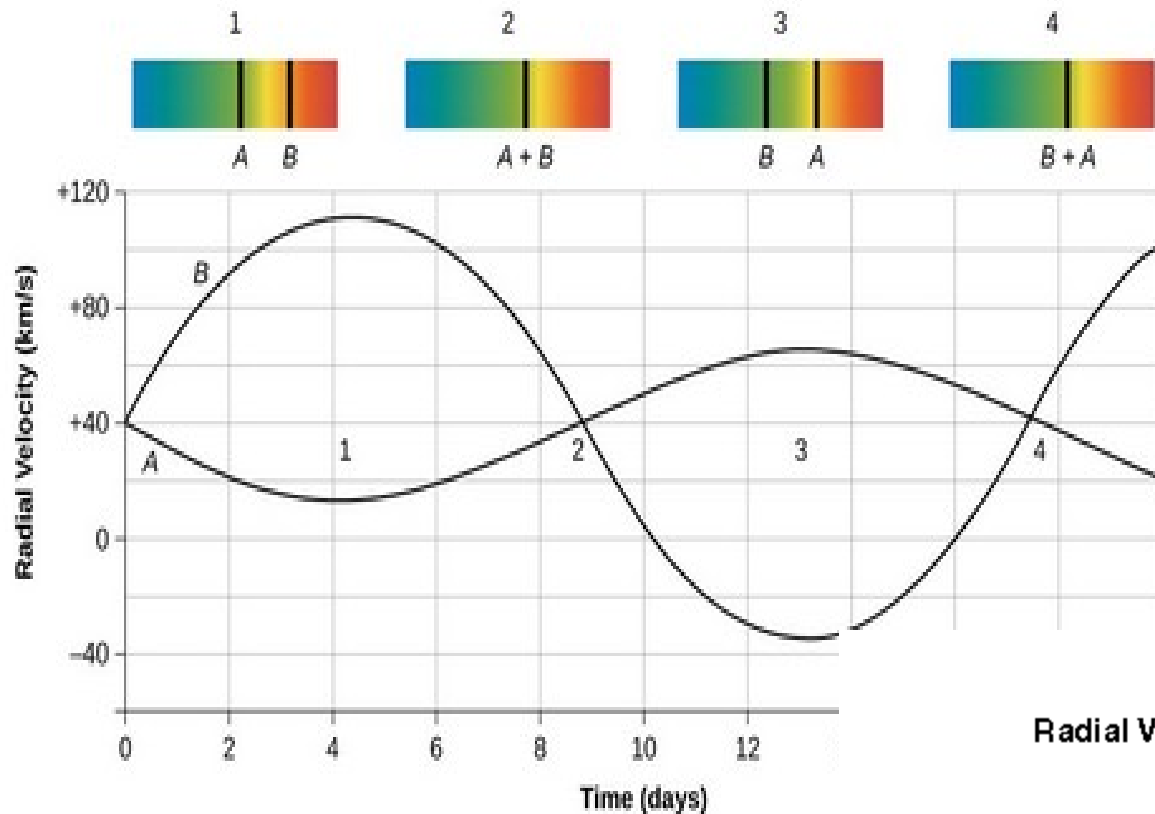


Law enforcement:
The frequency is
Shifted after it
Bounces off the
Police car. 15

DOPPLER SHIFT : The color of a star shift when it moves relative to us

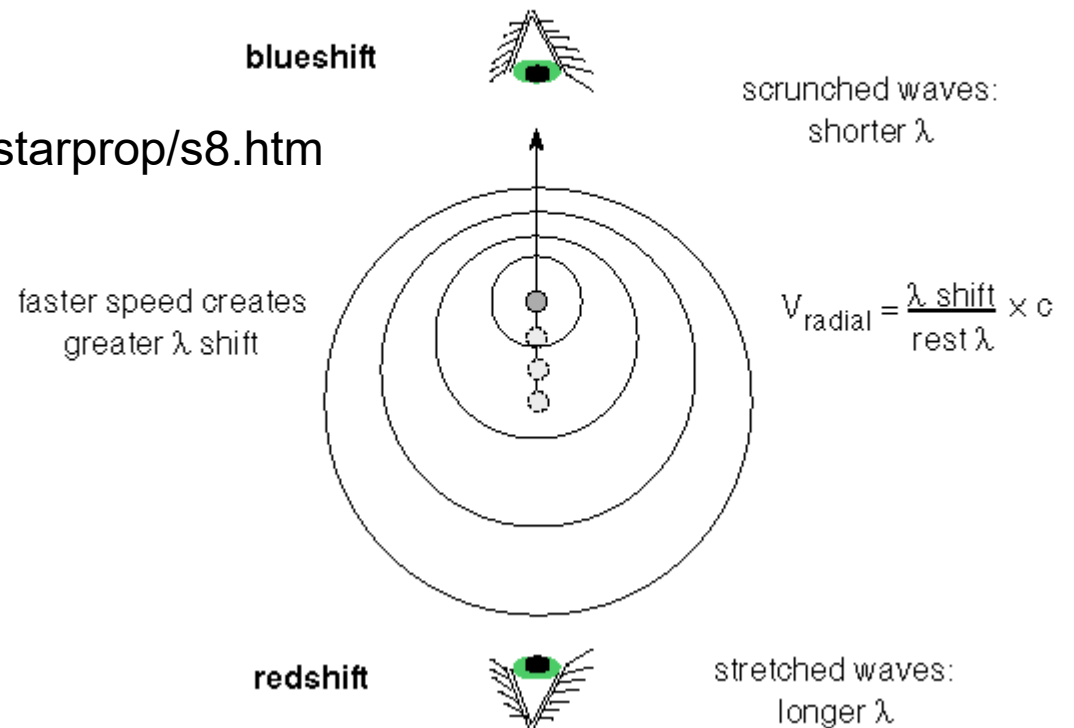


See other on my website

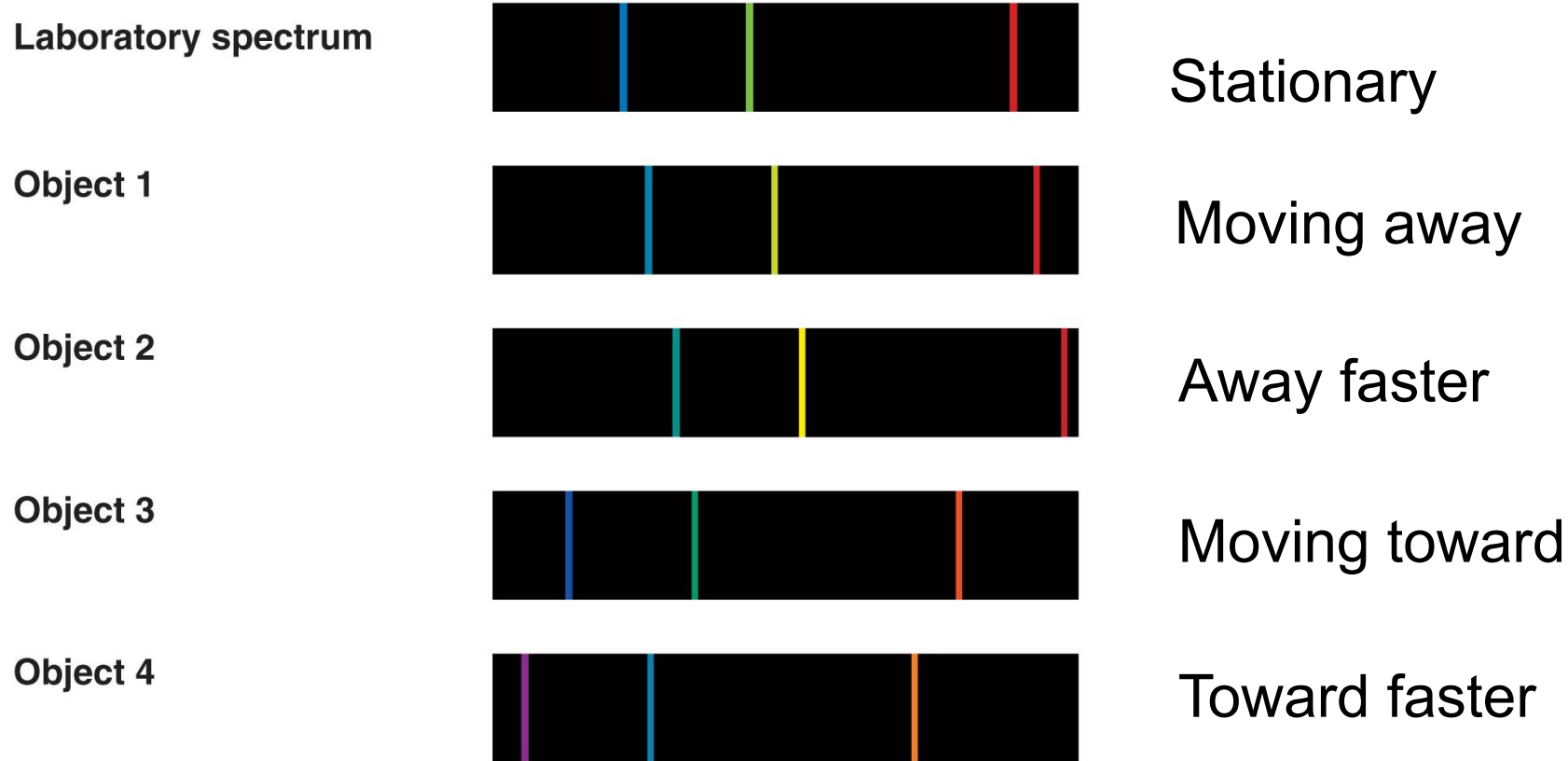


Radial Velocity measured from the *doppler effect*

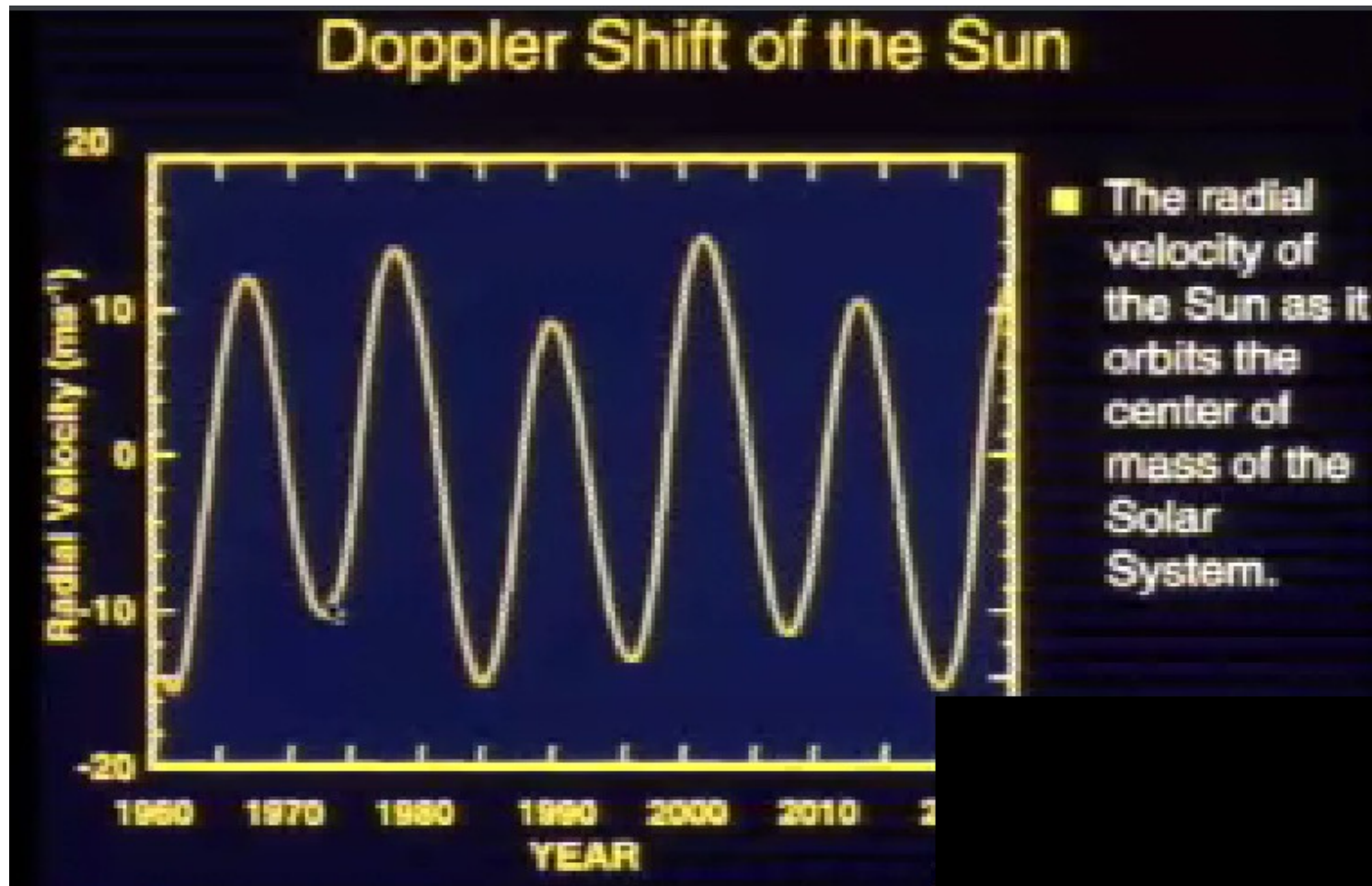
<https://www.astronomynotes.com/starprop/s8.htm>



Measuring the Shift



- We generally measure the Doppler effect from shifts in the wavelengths of spectral lines.

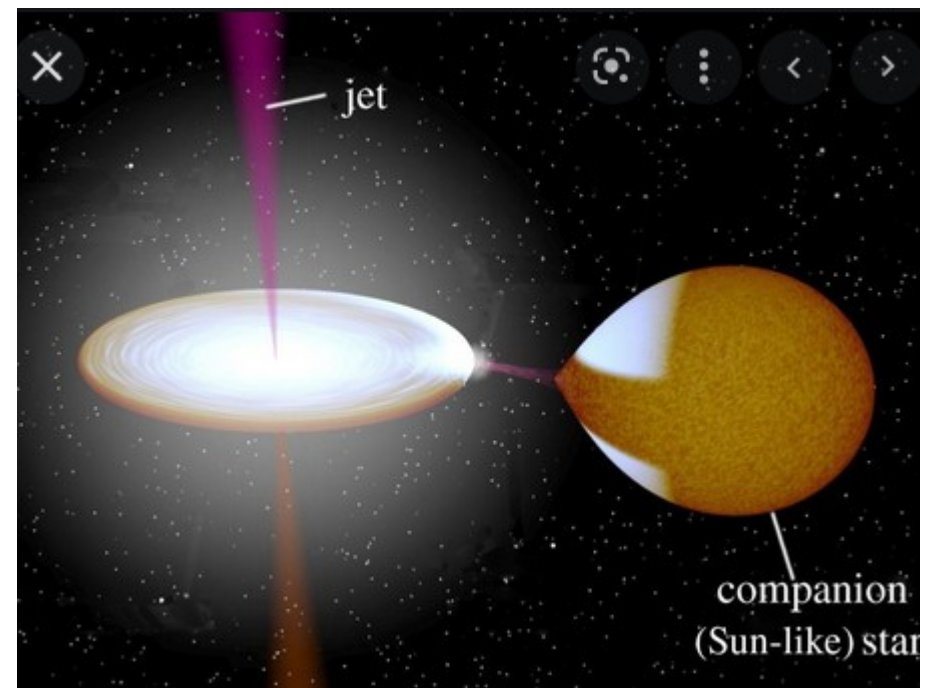


Doppler effect as seen by an alien aiming its telescope at the Sun's orbit around its Center of mass. (edge on). It will see a period of 12 years. That's the period of Jupiter. Sun and Jupiter orbit the same center of mass. Then it will also see another periodic Pattern with a period of 30 years, that's Saturn. Earth can't be detected. Only The wobble of the Sun due to the massive planets can be detectable. Note that we also get the speed about 12m/s .

This illustration depicts 4U 0614+091, an exotic binary system composed of two stellar remnants, a neutron star and a white dwarf (jets of X-rays)



<https://sci.esa.int/web/xmm-newton/-/47230-illustration-of-4u-0614-091>



Some stars like cuddling → contact binary star



giant star HR 5171 A

The European VLT telescope has studied the largest-known yellow star measuring more than 1,300 times the diameter of our sun. It is accompanied by another star which is so close that the two are touching and look like a peanut. HR 5171 A is establishing itself as an unusual star. Its gigantic size means that it shines one million times brighter than the Sun (hypergiant)



Algol, the demon star

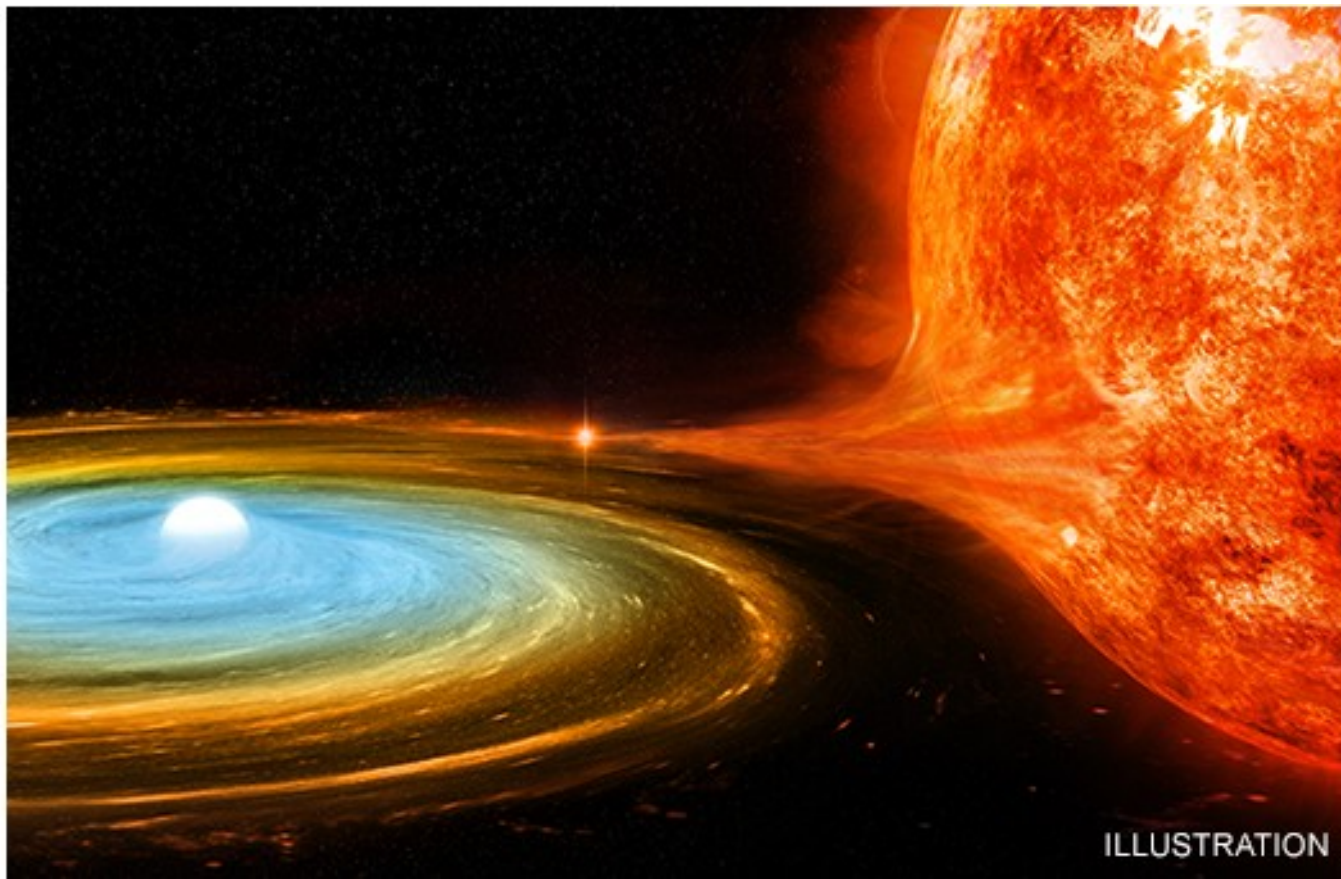


movie



Algol is a variable star that fluctuates in brightness from magnitude 2.1 to magnitude 3.4. Algol is actually a binary pair, with one star eclipsing the other every 2.867 days, with the actual drop in brightness lasting just a matter of hours.

<https://socratic.org/questions/what-is-algol-paradox-and-its-resolution>



Astronomers have detected a bright X-ray outburst from a star in the Small Magellanic Cloud, a nearby galaxy almost 200,000 light years from Earth. A combination of X-ray and optical data indicate that the source of this radiation is a white dwarf star that may be the fastest-growing white dwarf ever observed.

This type system → novae, or supernovae type Ia (standard candles)

Movie transfer science guy (2:22) and the cbs one first

<https://chandra.harvard.edu/blog/node/701>